

Benthick 3B

A solid-state iron and transistor bus processor, laid out in signal order.



Overview

Benthick 3B is a bus processor for tight low-end weight, fast transient control and dense glue, without woolly valve artefacts. Its stages run left to right across the panel in signal order: FRONT END, COMP 1, the TILT knob, COMP 2, the SUM, BUS COMP and OUTPUT.

The front end is either IRON, an asymmetric transformer core that works hardest on the lows, or CLASS-A, a symmetric transistor clip with a dense edge in the upper mids. Two compressors then share the work: COMP 1 is a slow window that lets the early transients through, COMP 2 a fast FET-style grabber. BALANCE blends the two paths, and a console-style VCA BUS COMP pulls them together.

Every stage has its own IN key. There is no drift: Benthick 3B is clean and repeatable.

Reach for it for

- Drum bus weight, smash and parallel crush
- Mix bus glue and polish
- Low-end weight on bass DI and kick (IRON)
- Bringing vocals forward with CLASS-A edge

- Widening a master with M/S drive and side saturation

Controls

1 FRONT END

IN	Switches the front end in or out.
IRON / CLASS-A	IRON (default): an asymmetric transformer core that works hardest on the lows below about 160 Hz, for even harmonics and weight, with a little magnetic hysteresis. CLASS-A: a symmetric transistor clip with odd harmonics and a dense edge in the upper mids. Switching between them crossfades, so it never clicks.
DRIVE	How hard the front end is pushed, 0 to 18 dB (4 dB by default). Small signals pass at unity, so DRIVE adds colour rather than level.
M/S	Off (L / R): both sides are driven equally. On (SIDES DRIVEN): the front end works in mid / side, and the sides are driven harder for a natural width.
S DRIVE	How much harder the sides are driven when M/S is on, 0 to 12 dB (4 dB by default).

2 COMP 1 (SLOW WINDOW)

IN	Switches COMP 1 in or out.
THRESH	-40 to 0 dB (-18 dB by default).
RATIO	1:1 to 10:1 (2:1 by default).
ATTACK	10 to 80 ms (40 ms by default). The slow attack lets the early transients through untouched.
RELEASE	50 ms to 1 s (300 ms by default).
HYST	Hysteresis, 0 to 12 dB (3 dB by default). Once the level goes over THRESH, the release holds back until it falls

	HYST dB below it, so the compressor never chatters round the threshold.
SC HPF	A high-pass filter on what all three compressors listen to, 30 to 300 Hz (90 Hz by default). It keeps the lows from making the compressors pump; the audio itself isn't filtered.
GR	COMP 1's gain reduction, 1 to 20 dB.

TILT

TILT	The big knob between COMP 1 and COMP 2: a seesaw around 700 Hz, -6 to +6 dB. Turning it right lifts the top and lowers the bottom; left does the opposite. It comes after COMP 1, so it shapes both what COMP 2 hears and what you hear. The plate under it shows the value while you turn it.
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3 COMP 2 (FET)

IN	Switches COMP 2 in or out. With COMP 2 out, the SUM section is out too.
4 / 8 / 12 / 20	The ratio (4:1 by default). Higher ratios have a harder knee.
ALL	Every ratio key in: 20:1 with no knee, faster attack and release and more grit from the gain cell - hard and aggressive. Click again to go back to 4:1.
THRESH	-40 to 0 dB (-14 dB by default).
ATTACK	20 to 800 microseconds (400 by default) - fast enough to grab the transients.
RELEASE	50 ms to 1.1 s (300 ms by default).
MAKEUP	COMP 2's level, 0 to 24 dB (2 dB by default).
GR	COMP 2's gain reduction, 1 to 20 dB.

SUM

BALANCE

Blends the two paths: towards 1, the COMP 1 path with its untouched transients (after TILT); towards 2, the clamped COMP 2 signal. The value reads as the two shares, 40 / 60 by default. It is the same signal through two paths, so they sum without phase problems.

4 BUS COMP

Console-style VCA glue that pulls COMP 1 and COMP 2 together. Its detector shares the SC HPF.

IN	Switches the bus compressor in or out.
2 / 4 / 10	The ratio (2:1 by default).
ATTACK	Stepped: 0.1, 0.3, 1, 3, 10 or 30 ms (10 ms by default).
RELEASE	Stepped: 0.1, 0.3, 0.6 or 1.2 s, or A (AUTO, the default). AUTO releases quickly after a transient and slowly once it has been working for a while.
THRESH	-30 to 0 dB (-18 dB by default).
MAKEUP	0 to 18 dB (5 dB by default).
GR	The bus compressor's gain reduction, 1 to 20 dB.

5 OUTPUT

WIDTH	Mid / side width, 0 to 200% (100% by default).
S SAT	Drives the sides into a second transistor saturation stage, 0 to 100%. Adds density and width to the edges of the image.
MIX	Wet / dry: 100% is fully processed; lower blends the unprocessed input back in, for parallel processing.
OUTPUT	Output level, -24 to +12 dB.

GR display	Shows the gain reduction of COMP 1, COMP 2 and BUS in dB.
IN / OUT meters	Input level and output level (left and right). Click any meter to clear the peaks.
LIMIT	The final stage. Off (default): a soft clip that protects the output at -0.3 dBFS. On (red LED): a lookahead brickwall limiter that holds the peaks at -0.3 dBFS (3 ms latency, compensated by REAPER).

Benthick 3B has no IN / OUT strip: the input trim, the input guard (on), the band saturation (off) and the output ceiling stay at their defaults.

The header

PRESETS	Opens the list of factory presets; click the display for the list too. The display shows PRESET nn/10 and the name, or Custom. Choosing a preset sets every stage (and puts all the IN keys in); OUTPUT and LIMIT are left as they are. The last entry resets everything to the defaults.
◀ / ▶	Step to the previous or next preset.
S / M / L	Panel size: small, medium or large.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

Bus Glue	The default settings: IRON at 4 dB, gentle COMP 1, COMP 2 at 4:1, 2:1 bus glue with AUTO release.
Iron Low End	IRON driven to 8 dB, a little darker TILT, slower compressors, bus makeup 7 dB.
Drum Smash	CLASS-A, TILT +2 dB, COMP 2 on ALL with a 20 microsecond attack, 4:1 bus comp, a little width and S SAT.

Vocal Forward	Light CLASS-A, SC HPF 150 Hz, TILT +1.5 dB, COMP 2 at 8:1 weighted to 70%.
Wide Master	M/S with 6 dB side drive, gentle compression, 130% width and 25% S SAT.
Parallel Crush	CLASS-A driven to 10 dB, COMP 2 on ALL crushing hard, 10:1 bus comp, blended at 60% MIX.
Mix Bus Polish	Light IRON with M/S, very gentle compression, 115% width and a touch of S SAT.
Bass DI Weight	IRON driven to 10 dB, SC HPF 60 Hz, TILT -2 dB, COMP 2 at 8:1.
Room Mic Grit	CLASS-A at 12 dB with M/S, TILT +3 dB, COMP 2 on ALL, 140% width and 40% S SAT at 70% MIX.
Gentle Master	Barely any drive, 1.3:1 COMP 1, light bus glue, 105% width.
Reset all to defaults	Every control back to its starting value, including OUTPUT and the hidden input and output settings.

Tips

- Use BALANCE as the main 'how much' control: towards 1 keeps the punch, towards 2 adds density.
- If the low end makes the compressors pump, raise SC HPF; the bass itself is not filtered.
- Switch stages out with their IN keys to hear what each one adds - the controls of a stage that is out are greyed.
- Keep an eye on the GR display: a few dB on each compressor is usually enough for glue.

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